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TECHNOLOGY

وزارة التعليم
جامعة الإمام
عبدالرحمن بن فيصل
كلية علوم الحاسب
وتقنية المعلومات

جامعة الإمام عبدالرحمن بن فيصل
IMAM ABDULRAHMAN BIN FAISAL UNIVERSITY

ARTI 308 – Machine Learning
Term 2 – 2025/2026

Project Proposal Form

- ✓ Group #: 4
- ✓ Members of the group: 6

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- ✓ Title of the Project :**Analyzing Dietary Habits Using Unsupervised Learning Techniques**

Abstract

Dietary habits have a direct impact on human health, yet understanding eating behavior patterns remains a complex analytical challenge due to the diversity and variability of nutritional data. This proposal aims to develop an unsupervised machine learning framework to analyze dietary habits by discovering hidden behavioral patterns without relying on labeled datasets, enabling the automatic identification of natural groupings of individuals based on food consumption and lifestyle characteristics. The motivation behind this work arises from the increasing availability of dietary and health-related data alongside the limitations of traditional analysis methods that depend on predefined classifications. Unsupervised learning provides an opportunity to explore data objectively and uncover patterns that may otherwise remain unnoticed. The proposed approach involves data preprocessing, feature extraction, and the application of clustering and dimensionality reduction techniques to reveal meaningful dietary structures. The expected outcome is a data-driven analytical model capable of identifying distinct dietary profiles and behavioral trends, supporting health researchers, nutrition specialists, and decision-makers in developing targeted nutritional strategies and improving dietary awareness initiatives.

Introduction

Dietary behavior is one of the most influential factors affecting public health, contributing to both disease prevention and overall well-being. With the rapid advancement of digital health systems and wearable technologies, large volumes of nutritional and lifestyle data are continuously being generated. Despite this growth, transforming raw dietary data into meaningful insights remains a significant challenge due to its complexity, variability, and lack of clear labeling. This proposal introduces an unsupervised dietary habit analysis framework designed to explore eating patterns using machine learning techniques.

The primary aim of this work is to analyze dietary data without predefined categories, allowing patterns to emerge naturally from the data. This approach addresses the limitations of manual classification and supervised learning models, which often require extensive labeled datasets and may introduce bias through predefined assumptions. By applying unsupervised learning methods, the proposed study seeks to uncover hidden relationships among food consumption behaviors, frequency patterns, and lifestyle indicators.

Previous studies have explored dietary analysis using statistical approaches and supervised machine learning techniques to predict health outcomes or classify nutritional behavior. While these efforts have provided valuable contributions, they typically depend on labeled datasets and predefined target variables, limiting their ability to discover unknown behavioral structures. Therefore, there remains a need for exploratory data-driven approaches capable of revealing new insights without prior labeling.

The proposed work will employ data preprocessing, feature engineering, clustering algorithms, and visualization methods to identify meaningful dietary patterns and behavioral groups. The envisioned outcome is a scalable analytical framework that enhances understanding of dietary habits and supports future research in personalized nutrition and health analytics.

Problem Statement

In today's world, dietary habits play a crucial role in determining an individual's overall health and well-being. However, with the vast amount of dietary data available—such as food consumption patterns, nutrient intake, and meal timings—identifying meaningful patterns and trends can be a challenging task. Traditional analysis often requires labeled data, which is not always available, especially when dealing with large-scale datasets collected from diverse populations.

The problem addressed in this project is the identification of inherent patterns and relationships within dietary habits using unsupervised learning techniques. By applying clustering and dimensionality reduction methods, we aim to uncover hidden structures within the data, such as groupings of similar dietary patterns, associations between certain foods, and correlations between dietary habits and health outcomes. These findings could help in providing personalized nutrition recommendations and improving public health strategies.

Motivation

Our motivation for this work can be stated as follows:

- The quest for higher precision in identifying complex behavioral patterns in public health data.
- The ability of unsupervised learning methods to detect natural clusters without labeled data.
- Increasing concern about unhealthy eating habits and related diseases.

Objectives

Our objectives of this work can be stated as follow:

- Collect and preprocess dietary habit data.
- Apply unsupervised learning algorithms for clustering dietary behaviors.
- To create visual charts that make it easy to see the differences between the dietary groups we find.

Limitation

Data Reliability: The study is limited by the accuracy of self-reported dietary data, which is prone to recall bias and under-reporting.

Temporal Scope: This research will analyze a "snapshot" of dietary habits (a specific 24-hour period or 1-week average) and will not cover long-term longitudinal changes in eating behavior.

Causal Inference: As an unsupervised learning project, the scope is restricted to identifying patterns and correlations; it will not attempt to prove direct causation between specific diets and chronic diseases.

Nutritional Focus: The analysis will focus primarily on macronutrients (carbohydrates, fats, proteins) and major food groups, leaving out micronutrient details (vitamins/minerals) due to data complexity.

Notations

The following notations and abbreviations will be used throughout this proposal:

- **ML:** Machine Learning
- **UL:** Unsupervised Learning
- **K**: Number of clusters in clustering algorithms
- **x_i**: Data point representing an individual's dietary record
- **C_k**: Cluster k, representing a group of similar dietary habits
- **μ_k:** Centroid (mean point) of cluster k
- **PCA**: Principal Component Analysis
- **t-SNE**: t-Distributed Stochastic Neighbor Embedding
- **DBI**: Davies-Bouldin Index
- **WSS**: Within-Cluster Sum of Squares (Inertia)
- **Silhouette Score**: Clustering evaluation metric

Literature Survey Outcome

Recent advancements in machine learning have enabled researchers to move beyond traditional dietary analysis. Studies such as [1], [2], and [3] have demonstrated the effectiveness of unsupervised learning in identifying dietary patterns linked to chronic diseases like diabetes and hypertension across different global populations. Additionally, study [4] applied unsupervised learning techniques to explore dietary intake data, further highlighting the ability of such methods to uncover hidden patterns without predefined labels. Furthermore, [5] utilized complex network-based algorithms to link dietary behaviors with cardiovascular risks, while [6] compared various unsupervised methods to better understand lifestyle behaviors in children.

Although these studies provide valuable insights, they are often limited in scope and generalizability. Most of the existing research focuses on disease-specific outcomes such as diabetes or obesity, or targets specific population groups such as the elderly or children. There is still a lack of studies that explore dietary habits in a general population context without relying on predefined health labels or assumptions.

Therefore, a clear research gap exists in applying unsupervised learning techniques to analyze broader and more diverse dietary data. This project aims to address this gap by applying clustering methods such as K-means along with dimensionality reduction techniques like PCA to uncover latent dietary patterns. The goal is to provide a purely data-driven classification of eating behaviors that can support personalized lifestyle recommendations, rather than focusing solely on clinical diagnosis.

Methodology

The methodology for this project will follow standard machine learning development procedures:

1. **Data Collection**

   Obtain dietary habit datasets from publicly available health or nutrition repositories.

2. **Data Preprocessing**

   Handle missing values, normalize numerical features, and encode categorical variables.

3. **Feature Selection**

   Select relevant attributes such as meal frequency, calorie intake, food categories, and nutrient values.

4. **Model Development**

   Apply at least two unsupervised learning techniques:

   - K-Means Clustering
   - Hierarchical Clustering

5. **Visualization**

   Use PCA or t-SNE for dimensionality reduction and cluster visualization.

6. **Interpretation**

   Analyze the resulting clusters to identify dietary habit groups.

Criteria for Performance Evaluation

Since the proposed work focuses on an unsupervised learning approach to analyze dietary habits, the performance cannot be measured using traditional labels (like Accuracy or F1-Score). Instead, the model's effectiveness will be evaluated based on its ability to produce distinct, cohesive, and interpretable clusters. The following criteria will be used for measurement and comparison:

- **Internal Validation Metrics:**
  - **Silhouette Coefficient:** This will be the primary metric to measure the quality of the clusters. It assesses how similar an object is to its own cluster compared to other clusters. A score closer to +1 indicates that the dietary patterns are well-separated.
  - **Davies-Bouldin Index:** This will be used to evaluate the average similarity between each cluster and its most similar one. Lower values will signify better separation and compactness of the dietary groups.
  - **Inertia (Within-Cluster Sum of Squares):** To measure how tightly the data points (individuals' food intake) are grouped around their respective cluster centroids.
- **Model Selection Criteria:**
  - **The Elbow Method:** This technique will be applied to justify the choice of the optimal number of clusters (K). By plotting the inertia against the number of clusters, we will identify the "elbow point" where adding more clusters no longer significantly improves the model's performance.
- **Visual and Qualitative Assessment:**
  - **Dimensionality Reduction (PCA/t-SNE):** To compare the performance of different algorithms (K-Means vs. DBSCAN), visualization will be used to see how clearly the dietary habits are partitioned in a 2D/3D space.
  - **Cluster Interpretability:** Following the literature ([1], [6]), the success of the model will also be measured by the "meaningfulness" of the clusters. A high-performing model must produce groups that can be clearly labeled (High-Protein Diet, Fast-Food Dominant) to provide actionable nutritional insights.

Data

The dataset used in this project will be obtained from the National Health and Nutrition Examination Survey (NHANES), which is publicly available through the Centers for Disease Control and Prevention (CDC). The dataset can be accessed from the official NHANES website.

The NHANES dataset provides detailed information about individuals' dietary intake, nutritional values, and health-related variables collected from a large and diverse population.

The dataset includes several important features relevant to this study, such as:

- Daily calorie intake
- Macronutrient values (carbohydrates, proteins, fats)
- Food categories and meal types
- Frequency of food consumption
- Demographic information (age, gender)
- Lifestyle-related attributes (physical activity level, BMI)

- The dataset will be downloaded in CSV format and processed using Python.

Tools

Mention the software or hardware tools that shall be used for the proposed research.

Example could be: to implement the proposed models, several codes will have to be written. All coding will be done using Python, MATLAB or any other. Though, some modules may also be written in C++, to be called from MATLAB, if necessary. Weka, python, etc are all possible options

Programming Language: All core coding and data manipulation will be performed using Python.

Data Libraries: Pandas and NumPy will be used for preprocessing high-dimensional dietary tables and matrix operations.

Machine Learning Frameworks: Scikit-Learn will be utilized to implement clustering algorithms (K-Means, GMM) and dimensionality reduction (PCA).

Visualization Tools: Matplotlib and Seaborn will be used to generate performance evaluation charts, such as Silhouette plots and the Elbow method.

Development Environment: Jupyter Notebooks or Google Colab will serve as the primary IDE for iterative development

Planning

The research plan is summarized in the Gantt chart shown below. The project will be carried out according to the official schedule provided in the course description to ensure timely completion of all deliverables.

The first phase involved project topic selection and initial brainstorming, which was completed before the proposal submission deadline. The current phase focuses on preparing and submitting the project proposal, including defining the problem, reviewing the literature, and identifying the methodology.

Following proposal approval, the team will proceed with dataset selection and preprocessing, including cleaning, handling missing values, and preparing the data for analysis.

Next, unsupervised learning techniques such as K-means clustering and Principal Component Analysis (PCA) will be implemented and tested to analyze dietary habits and extract meaningful patterns.

After that, model evaluation and result analysis will be conducted, followed by the preparation and submission of the progress report.

In the final phase, the team will complete the final report, including all experimental results, analysis, and references. The project will conclude with a formal presentation of the findings.

PROJECT TIMELINE:

Task	Before Feb 19	Feb 19 – Apr 2	Apr 3 – May 7	May 8 – June 2	June 2 – 4
Project Selection	✓				
Proposal Preparation		✓			
Data Collection & Preprocessing			✓		
Model Implementation (K-means, PCA)			✓		
Evaluation & Analysis			✓		
Progress Report Submission			✓		
Final Report Preparation				✓	
Final Presentation					✓

Contributions

- Supporting nutrition-related research.
- Providing insights into personalized dietary recommendations.

Possible publication(s) from the research outcomes

The following are part of the envisaged possible publications during the course of the research.

Conference Paper: A submission to a regional technical conference, such as the *IEEE Saudi Arabia Sub-Section Conference*, focusing on the AI methodology applied to nutritional data.

Journal Article: A manuscript targeted at an interdisciplinary journal like *Nutrients (MDPI)* or *Frontiers in Nutrition*, highlighting the discovery of new dietary patterns in specific populations.

University Symposium: A poster presentation at the *IAU College of Computer Science & Information Technology Annual Research Forum*.

Conclusion

This project aims to bridge the gap between complex nutritional data and actionable health insights by developing a robust unsupervised machine learning framework. By shifting away from traditional, bias-prone labeled datasets and predefined dietary classifications, this research will allow natural eating patterns to emerge objectively through K-Means clustering and Hierarchical Clustering.The integration of dimensionality reduction techniques, such as PCA and t-SNE, will provide the necessary visual evidence to distinguish between diverse dietary profiles, while rigorous evaluation using the Silhouette Coefficient and Davies-Bouldin Index will ensure the mathematical reliability of the discovered patterns. As outlined in the project timeline, the transition from data preprocessing to model implementation will culminate in a scalable analytical model by June 2026.Ultimately, this work contributes to the advancement of personalized nutrition and public health analytics. By identifying latent behavioral structures within the NHANES dataset, the project provides a data-driven foundation for future research in chronic disease prevention and health awareness, supporting both regional academic forums and global interdisciplinary publications.

References

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[2] L. Li, H. Momma, H. Chen, and Y. Xu, "Dietary patterns associated with the incidence of hypertension among adult Japanese males: application of machine learning to a cohort study," European Journal of Nutrition, 2024.

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[5] M. Côté, J. M. Hutchinson, M. Touvier, and B. Srour, "Associations between dietary pattern networks derived from machine learning algorithms and cardiovascular risk in the NutriNet-Santé cohort," The Journal of Nutrition, 2025.

[6] N. J. D'Souza, K. Downing, G. Abbott, and L. Orellana, "A comparison of children's diet and movement behaviour patterns derived from three unsupervised multivariate methods," PLoS ONE, 2021.